Exercise-policy loss under computational and financial constraints

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Abstract

An approximate exercise policy can be difficult to assess because its decisions are suboptimal, because the upper bound used to assess them is loose, or because the financial comparison assigns different exercise opportunities. I develop a framework that separates these objects in a finite-calendar stopping problem with deterministic funding and discrete cash events. A martingale decomposition distinguishes expected policy loss from dual slack. Analytically centred piecewise-linear values provide a practical upper construction. Nested calendars reduce a prescribed financial and expiry sensitivity family to its maximal endpoint for unchanged exercise instructions. A conditional terminal adjustment supplies an expected policy gain that can be isolated with a common upper payoff. A predictable-compensation argument quantifies how conditional drift would affect loss bounds and paired policy comparisons. A repeated-training variance decomposition separates training variation from evaluation noise when preceding evidence determines fixed sample counts. Independent evaluation, simultaneous scenario limits and timing accounting delimit the resulting inference. In this methods-only version, I present the model, derivations and experimental design. I report no market observations, fitted empirical quantities or numerical research findings.

Keywords: optimal stopping; American options; martingale duality; Monte Carlo; discrete dividends; computational allocation.

1 Introduction

The economic value of an exercise algorithm is the expected reward obtained by its stopping rule. Regression fit, numerical resolution and the number of simulated paths are means of constructing or evaluating that rule; none is itself a measure of value. A useful comparison must identify the financial problem, specify a feasible policy and state what its benchmark establishes. A finely resolved price can be a valuable numerical reference without being a known optimum. A valid upper bound can remain too loose to establish that policy loss is economically small.

I ask three related questions. How much expected value does a fixed exercise instruction forgo within a specified model? How much computation is needed to measure that loss with useful precision? How does the answer change when the financial inputs or opportunity set change? These questions require different comparisons. Changing the martingale used for evaluation changes bound tightness. Changing a policy changes its expected reward. Changing the exercise calendar can change the optimum even when the original policy is left untouched.

The methodological foundations are established. Longstaff and Schwartz (2001) construct exercise policies through backward continuation regression. Haugh and Kogan (2004) develop martingale upper bounds and the use of approximate value processes. The lognormal hinge expectation used below follows the analytical pricing foundation of Black and Scholes (1973). I apply these foundations to an explicitly funded cash-event model and organise the resulting policy comparisons. I do not claim the dual inequality, conditional Jensen argument or finite-family inference as new general theorems.

Pricing intervals and policy assessment are already central to Andersen and Broadie (2004). Avoiding nested simulation is also established: Belomestny et al. (2009) construct martingale-preserving estimators, and Cline (2026) studies regression-based duals with explicit conditional restrictions. Consequently, analytical centring and lower computational cost do not by themselves identify a new contribution. I use the funded cash-event setting to separate changes in policy reward, evaluation slack and exercise opportunities. A claim about their economic importance requires controlled evidence on each margin.

Appendix A applies predictable compensation to penalties with conditional drift. It gives a loss allowance and a paired-comparison correction, together with a finite analytical example. Appendix B specifies the information available to a learned policy and derives the variance of a repeated-training comparison when earlier evidence determines its sample sizes. I present these as consequences of established theory. The literature comparison does not establish priority for the propositions or their proofs.

I restrict this mathematical component of my research to a framework independent of any particular observed contract, fitted parameter, sample size or empirical result. I specify the evidence required for numerical and financial investigation; I make no claim here that those empirical requirements have been met.

2 The funded finite-calendar stopping problem

2.1 State, payments and information

Consider a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{Q})$ carrying a Brownian motion. Let $D_0(u) > 0$ denote a deterministic discount factor for a payment at date u. If a quoted share price S_t settles at $u(t)$, define

$$Y_t = f_t S_t, \qquad f_t = D_0(u(t)). \tag{1}$$

The variable Y represents discounted economic ownership under the maintained settlement convention. This transformation requires the share and exercise transfer to represent matching economic entitlements. It does not establish a financing strategy or identify the correct funding curve from market prices alone.

Between prescribed cash events, I assume

$$dY_t = \sigma Y_t \, dW_t^{\mathbb{Q}}, \qquad \sigma \geq 0. \tag{2}$$

At an event with proposed discounted cash $c \geq 0$, the detached entitlement and remaining state are

$$C = \min(Y_{t-}, c), \qquad Y_t = (Y_{t-} - c)^+, \qquad Y_t + C = Y_{t-}. \tag{3}$$

The cap preserves non-negativity and cash accounting. This bounded cash specification is related to the stock-dependent dividend framework discussed by Vellekoop and Nieuwenhuis (2006); the settlement and ownership equations here specify the particular maintained model. The ex-entitlement state is not a martingale across the cash detachment; between events it is. Conditional transition calculations must respect this distinction. Proposed cash, volatility and funding are model inputs, not identified expectations merely because they permit a price fit.

For a put with strike $K > 0$, let $\mathcal{E} = \{t_1, \ldots, t_J\}$, with $0 < t_1 < \cdots < t_J$, be a finite set of admissible exercise rights. Valuation is at $t_0 = 0$. Define

$$H_j = \left(K f_{t_j} - Y_{t_j}\right)^+. \tag{4}$$

Write $Y_j = Y_{t_j}$ and $M_j = M_{t_j}$ when processes are indexed by the decision grid; $H_{t_j} = H_j$ identifies the reward at a calendar time. Thus H_τ denotes the reward at the selected exercise right. The exercise-settlement factor can differ from the premium-payment factor $b > 0$. Values below are in discounted currency unless divided by b. The definition of H_j presumes that the

specified exercise settlement transfers the intended ownership state and entitlements. A change in deliverable or exercise funding can require a different reward.

Assumption 1 (Admissibility)**.** *The decision set is finite and fixed before evaluation. Rewards are adapted and integrable. Policies are stopping times taking values in $\mathcal{E}$. Evaluation martingales start at zero and are integrable at all rights. Whenever sampling variances are reported, the corresponding random variables have finite second moments. Learned coefficients and approximate value functions are fixed before independent evaluation paths are drawn.*

With deterministic initial state and model inputs, the optimal and policy values are

$$V^* = \sup_{\tau \in \mathcal{T}(\mathcal{E})} \mathbb{E}[H_\tau], \qquad V_\pi = \mathbb{E}[H_{\tau_\pi}], \qquad \ell_\pi = \frac{V^* - V_\pi}{b} \geq 0. \tag{5}$$

Cash-event observations needed for propagation do not automatically grant exercise rights. The order of an exercise decision, detachment and settlement is part of the problem definition. A finite exercise calendar represents a Bermudan problem; its numerical refinement alone does not verify the contractual exercise opportunities of an American option.

2.2 Learned exercise instructions

Backward regression estimates continuation values on a training sample and converts them into stopping predicates (Longstaff and Schwartz, 2001). All coefficients, basis choices, positivity safeguards and tie rules form part of the frozen policy. Independent forward evaluation estimates its value without treating its training fit as realised out-of-sample accuracy. Increasing basis dimension need not improve a finite-sample rule.

When a financial scenario θ changes the ownership factor, I preserve the instruction in quoted-price coordinates. If a_j^θ is the scenario share-settlement factor, the quoted state is $Q_j = Y_j^\theta / a_j^\theta$. A source predicate expressed in its original ownership coordinate receives

$$T_j^\theta(Y_j^\theta) = Y_j^\theta \frac{a_j^0}{a_j^\theta}. \tag{6}$$

Its original coefficients and decision conventions are retained, but the resulting stopping time is valued with scenario rewards. This preserves the instruction for an identical history of quoted prices. It does not preserve the distribution of stopping dates when the scenario dynamics change.

3 Loss measurement and dual slack

3.1 A pathwise gap

For an admissible martingale M with $M_0 = 0$, set

$$U(M) = \max_{1 \leq j \leq J} \{H_j - M_j\}, \qquad L_\pi(M) = H_{\tau_\pi} - M_{\tau_\pi}, \qquad G_\pi(M) = U(M) - L_\pi(M). \tag{7}$$

Here the maximum ranges over the indexed exercise rights. The martingale dual approach is due to the published framework of Haugh and Kogan (2004).

Proposition 1 (What a gap measures)**.** *Under Assumption 1, $G_\pi(M) \geq 0$ pathwise and*

$$\mathbb{E}[G_\pi(M)] = \underbrace{V^* - V_\pi}_{\textit{policy loss}} + \underbrace{\mathbb{E}[U(M)] - V^*}_{\textit{dual slack}}. \tag{8}$$

Both terms on the right are non-negative.

Proof. The maximum defining U includes the policy's stopping right, proving the pathwise inequality. Finite-horizon optional sampling gives $\mathbb{E}[M_\tau] = 0$ for each admissible stopping time. Hence $\mathbb{E}[L_\pi] = V_\pi$. Also $U \geq H_\tau - M_\tau$ for every such τ, so $\mathbb{E}[U] \geq \sup_\tau \mathbb{E}[H_\tau] = V^*$. Subtracting V_π and adding and subtracting V^* gives (8). □

For a fixed policy and financial problem, a change in expected gap caused only by replacing M is a change in dual slack. It is not a change in policy value. A numerical PDE value supplies an additional comparison but is not inserted as an exact V^* without a justified error enclosure. Mesh agreement is evidence about numerical behaviour, not a proof of the financial model.

If two policies use the same upper payoff and martingale, then

$$G_\pi(M) - G_\rho(M) = L_\rho(M) - L_\pi(M), \qquad \mathbb{E}[G_\pi - G_\rho] = V_\rho - V_\pi. \tag{9}$$

The upper payoff cancels pathwise. With the martingale condition satisfied, paired gap differences identify expected policy gains without confounding them with changes in the bound. Appendix A gives the additional compensator term when that condition fails.

3.2 Analytically centred approximate values

Let v_{j+1} be a deterministic approximation to a subsequent value function, or a function fixed by an independent training procedure. Define

$$\Delta M_{j+1} = v_{j+1}(Y_{j+1}) - \mathbb{E}[v_{j+1}(Y_{j+1}) \mid \mathcal{F}_{t_j}]. \tag{10}$$

When the conditional expectation is evaluated for the actual retained function and transition, these increments have conditional mean zero in exact arithmetic. Approximation quality affects tightness rather than supplying the reason for centring. The construction is the martingale component of an approximate value process described by Haugh and Kogan (2004).

I represent approximate values on the non-negative state domain as

$$v(y) = \alpha + \beta y + \sum_{r=1}^{R} \gamma_r \left(y - k_r\right)^+, \qquad k_r \geq 0. \tag{11}$$

Over a no-cash interval of length h, $X = y \exp(-\sigma^2 h/2 + \sigma\sqrt{h}\, Z)$, with Z standard normal. For $y, k > 0$ and $\sigma^2 h > 0$,

$$\mathbb{E}[(X-k)^+] = y\Phi(d_1) - k\Phi(d_2), \tag{12}$$

$$d_1 = \frac{\log(y/k) + \sigma^2 h/2}{\sigma\sqrt{h}}, \qquad d_2 = d_1 - \sigma\sqrt{h}. \tag{13}$$

This is the lognormal option integral (Black and Scholes, 1973). Zero state, strike, variance or time uses its continuous limiting form. Each term in (11) can consequently be integrated over the full support, including the retained tail rule.

For a capped cash jump with size c, composition is explicit:

$$v((x-c)^+) = \alpha + \beta\,(x-c)^+ + \sum_{r=1}^{R} \gamma_r \left(x - (c + k_r)\right)^+. \tag{14}$$

The identity uses $k_r \geq 0$ and includes the constant segment for $x \leq c$. Cash-phase ordering must be preserved when multiple propagation intervals are joined. An approximate quadrature expectation or truncated integration range cannot simply be substituted into (10) while retaining an exact martingale claim.

Floating-point evaluation, function compression and approximation of the dynamic programme are distinct numerical errors. Validation should check analytical limiting cases, cash accounting, tail treatment and independent pricing references. A compression tolerance for one function is not an accumulated error bound for the stopping problem.

4 Changed opportunities and changed policies

4.1 Nested calendars with unchanged instructions

Let $\theta \in \Theta$ index a prescribed finite financial family and let $\mathcal{E}(d)$ be nested exercise calendars. Write $V_\theta^*(d)$ for the optimum and suppose that the original policy always stops on the smallest calendar. Its reward definition and pre-terminal dynamics remain unchanged when d varies within a financial scenario.

Proposition 2 (Maximal-calendar reduction)**.** *If $\mathcal{E}(d_1) \subseteq \mathcal{E}(d_2)$ whenever $d_1 \leq d_2$, and $b_\theta > 0$ is constant across these calendars, then*

$$\max_{\theta\in\Theta,\, d\in D} \frac{V_\theta^*(d) - V_{\pi,\theta}}{b_\theta} = \max_{\theta\in\Theta} \frac{V_\theta^*(d_{\max}) - V_{\pi,\theta}}{b_\theta}. \tag{15}$$

Proof. Every stopping rule on a smaller calendar is feasible on a larger one, so the optimum is nondecreasing. The original policy value is constant within the scenario because its decisions and rewards remain unchanged. Subtracting this constant and dividing by a positive fixed premium factor preserves monotonicity. Taking the maxima gives (15). □

The statement concerns true expected losses. Sample upper limits need not increase with the calendar because sampling noise and dual slack can change. Evaluating the prespecified maximal calendar supplies a bound for smaller nested calendars; it does not claim to have priced or simulated every intermediate case. Recalibration across endpoints, a changed cash transfer or a policy allowed to use new observations would require a different comparison.

4.2 A feasible terminal adjustment

Suppose the original endpoint is T, the extended endpoint is $T + h$, and between them Y is an integrable martingale with no cash event and a constant exercise threshold k. Define

$$g(y) = (k - y)^+, \qquad g_h(y) = \mathbb{E}[g(Y_{T+h}) \mid Y_T = y], \qquad q_h(y) = g_h(y) - g(y). \tag{16}$$

Conditional Jensen gives $q_h(y) \geq 0$. Under the lognormal transition, g_h follows the corresponding analytical put integral. The endpoint condition is stronger than calendar nesting: it provides a tractable terminal replacement and dominance of earlier exercise within the added interval.

Let $A = \{\tau_\pi = T\} \in \mathcal{F}_T$. Change only the final instruction: if A occurs, wait until $T + h$; otherwise retain the original stopping time. Denote this adjusted policy by $\widetilde{\pi}$.

Proposition 3 (Expected gain from the final instruction)**.** *The adjusted rule is feasible on the expanded calendar and*

$$V_{\widetilde{\pi}} - V_\pi = \mathbb{E}[\mathbf{1}_A q_h(Y_T)] \geq 0. \tag{17}$$

The conditionally integrated gain has variance no greater than the direct realised payoff difference, when the latter has finite variance.

Proof. The decision to wait is made using the $\mathcal{F}_T$-measurable event A, so the rule is a stopping time. Outside A, payoffs agree. On A, the tower property gives the expected difference as $q_h(Y_T)$. Conditional Jensen proves its sign. The integrated gain is the conditional expectation of the realised difference given $\mathcal{F}_T$; the law of total variance proves the variance comparison. □

The realised later payoff can still be smaller. Equation (17) establishes an expected improvement under the maintained model, not a pathwise profit or executable market opportunity.

For the expanded problem, a common upper payoff isolates this gain as in (9). The residual discounted loss satisfies

$$V_h^* - V_{\widetilde{\pi}} = (V_0^* - V_\pi) + (V_h^* - V_0^*) - \mathbb{E}[\mathbf{1}_A q_h(Y_T)]. \tag{18}$$

The policy gain need not be bounded above by the increase in the optimum: a suboptimal original rule may survive where an optimal original rule would already stop. Residual loss for a changing policy need not be monotone in the endpoint.

4.3 Retraining as a separate comparison

Replacing the terminal training response by g_h can also change earlier continuation regressions. To isolate that effect, I use the same training paths, basis, regression safeguards and financial inputs for original and revised fits, and give both policies the same final waiting instruction. I freeze the fits before fresh paired evaluation.

Equation (17) does not imply that finite-sample retraining improves value. Changed stopping decisions alter the responses entering earlier regressions, and estimated continuation can move in either direction. A common upper payoff identifies the incremental expected value through (9). Unfavourable and inconclusive paired outcomes must remain in the comparison.

5 Conditional inference and scenario families

5.1 What the probability statement conditions on

With the financial model, fitted policy and approximate values fixed, simulate independent outer paths. Let $X_i = G_\pi^{(i)}(M)/b$, with sample mean $\overline{X}$ and sample standard deviation s_X. A commonly used approximate upper limit is

$$B_\pi = \overline{X} + t_{1-\alpha,n-1}\frac{s_X}{\sqrt{n}}. \tag{19}$$

Since $\ell_\pi \le \mathbb{E}[X_i]$, a valid upper limit for the expected gap is also an upper limit for policy loss. Student calibration is generally approximate for non-normal gaps; it is not a finite-sample guarantee. Finite variance, independent evaluation and sufficiently informative Monte Carlo samples remain necessary. More outer paths improve precision about the expected gap but do not remove its dual slack.

A conditional interval does not integrate over the uncertainty of policy training, financial calibration, contract conventions or sample selection. Training repetitions must be retained separately if their variation is investigated. Choosing a policy on the same evaluation outcomes used to advertise its bound can invalidate the selected-policy interpretation. Such a selection requires a separate evaluation design or an appropriate simultaneous guarantee.

5.2 Coverage for a prescribed financial family

For $m = |\Theta|$, compute a per-scenario limit at the maximal calendar using error allocation α/m:

$$B_\theta = \overline{X}_\theta + t_{1-\alpha/m,n_\theta-1}\frac{s_\theta}{\sqrt{n_\theta}}. \tag{20}$$

If each marginal limit has its stated coverage, the union bound and (15) imply an event of probability at least $1-\alpha$ on which every prescribed scenario and smaller calendar is covered. With approximate marginal calibration, the joint statement inherits that approximation. Independence across scenarios is unnecessary for the union bound.

There is no extra multiplicity factor for calendars already dominated by the maximal endpoint. There is also no automatic extension to all policies, a continuous funding region or scenarios introduced after inspection. A missing required scenario prevents a complete-family conclusion; the family size must not shrink opportunistically. Financial quote compatibility is a different condition and must be reported separately from mathematical completeness of the conditional family.

5.3 Pairing, independent runs and gap interactions

Within a common-path experiment, I estimate policy gains or scenario effects from pathwise differences. For $D_i = X_i - Y_i$,

$$\widehat{\mathrm{Var}}(\overline{D}) = \frac{s_D^2}{n} = \frac{s_X^2 + s_Y^2 - 2s_{XY}}{n}. \tag{21}$$

Discarding covariance would answer a different uncertainty question. Conversely, unrelated evaluation streams are not paired merely because policies or model labels coincide.

For independent mean estimates R and L, possibly with different path counts,

$$\widehat{v} = \frac{s_R^2}{n_R} + \frac{s_L^2}{n_L}, \qquad \nu = \frac{\widehat{v}^2}{(s_R^2/n_R)^2/(n_R-1) + (s_L^2/n_L)^2/(n_L-1)}. \tag{22}$$

The usual Welch interval uses $t_{1-\alpha/2,\nu}\sqrt{\widehat{v}}$. Degenerate zero-variance cases require explicit treatment. A difference between reported confidence endpoints is a descriptive comparison, not itself a confidence interval for a policy-value difference.

A financial/calendar interaction can be written in terms of expected quoted gaps as

$$I_\theta = \{g_\theta(d_{\max}) - g_0(d_{\max})\} - \{g_\theta(0) - g_0(0)\}. \tag{23}$$

When each financial effect is paired internally but the two calendar experiments are independent, add the variances of their paired mean effects. By (8), I_θ combines interactions in true loss and dual slack. It does not identify either component alone and has no causal interpretation. Repeated dependent pointwise comparisons do not constitute a simultaneous test; intervals containing zero do not establish equivalence or exact additivity.

6 Computational expenditure as part of the design

6.1 Definition of the computational workload

A time comparison is meaningful only after defining the computational unit. For a collection of frozen policies under one financial model, approximate-value construction and the common martingale can be shared. Charging this construction once per policy overstates its cost; omitting it understates the cost. A single-policy workload can have a different ranking from a repeated multi-policy workload.

The timed unit should state whether it includes simulation, deterministic approximation, conditional integration, martingale evaluation, policy realisation, estimator construction and numerical checks. Previous training, imports, data loading and file output should be reported separately if excluded. Warm-up and allocation pilots are additional expenses, not free inputs to an end-to-end valuation. Record elapsed and CPU time and the relevant hardware and thread configuration.

6.2 Path allocation before evaluation

I allocate paths using median pilot times t_1, t_2 at two path counts $n_1 < n_2$:

$$b_T = \frac{t_2 - t_1}{n_2 - n_1}, \qquad a_T = \max(0, t_1 - b_T n_1), \qquad \widehat{T}(n) = a_T + b_T n. \tag{24}$$

For target B and path increment q, choose

$$n(B) = q \left\lfloor \frac{B - a_T}{b_T q} \right\rfloor, \tag{25}$$

subject to prescribed limits. A non-positive fitted slope or infeasible minimum is a design failure. The unclipped intercept, timing dispersion and any cap must be retained. I fix all allocations before drawing the independent evaluation streams and retain each prescribed run irrespective of whether its confidence bound becomes favourable.

The affine model predicts time; it does not enforce it. Extrapolation beyond the pilot range, memory effects and host variability can create substantial discrepancies. I compare actual time with target time and retain overruns. A strict equal-cost ranking requires adequate timing calibration or a separately justified comparison on an observed cost frontier. Nominally equal targets alone do not establish it.

6.3 Allocating repetitions of a fixed training comparison

The value of a realised policy and the mean value of a training procedure are different estimands. To study the latter, I fix the financial specification, two training procedures, their training sample sizes and the common evaluation martingale. Let D contain the inputs needed to fit one pair of policies, with a prescribed joint law ν. This law may couple the fits through shared training paths, as in the retraining comparison above. Independence is required between successive pairs and their fresh evaluation paths, not between the two fits within a pair. Increasing the number of pairs repeats the same procedures; it does not enlarge the training sample of an individual fit.

For a complete evaluation path X, define the quoted adjusted gain

$$Z(D,X)=\frac{L_{\pi_1(D)}(M)-L_{\pi_0(D)}(M)}{b},\qquad \delta(D)=\mathbb{E}[Z\mid D],\qquad \sigma_D^2=\operatorname{Var}(Z\mid D). \tag{26}$$

Expectations in this subsection include both training and evaluation randomness. Under independent evaluation and the martingale condition, $\delta(D)=\{V_{\pi_1(D)}-V_{\pi_0(D)}\}/b$ almost surely. Assume Z has a finite second moment under the joint sampling law. The fixed target and its two variance components are

$$\mu=\mathbb{E}[\delta(D)],\qquad \mathcal{V}_{\mathrm{tr}}=\operatorname{Var}\{\delta(D)\},\qquad \mathcal{V}_{\mathrm{ev}}=\mathbb{E}[\sigma_D^2]. \tag{27}$$

Let $\mathcal{P}$ contain all preceding evidence used to choose positive integer counts m and n, before any new outcome is drawn. For each of m independent training pairs, I average n fresh evaluation gains to obtain W_r and then set $\widehat{\mu}=m^{-1}\sum_{r=1}^{m}W_r$. The product-space construction in Appendix B gives

$$\mathbb{E}[\widehat{\mu}\mid\mathcal{P}]=\mu,\qquad \operatorname{Var}(\widehat{\mu}\mid\mathcal{P})=\frac{\mathcal{V}_{\mathrm{tr}}}{m}+\frac{\mathcal{V}_{\mathrm{ev}}}{mn}. \tag{28}$$

I obtain this decomposition by conditioning first on the fitted pair and then averaging over independent pairs. Its purpose is to distinguish variation between fitted policies from the Monte Carlo error remaining after those policies are fixed. Additional paths within a pair reduce only the second component.

For an illustrative deterministic cost $C=c_0+m(c_{\mathrm{tr}}+c_{\mathrm{ev}}n)$, the unconstrained continuous relaxation gives

$$n^*=\sqrt{\frac{\mathcal{V}_{\mathrm{ev}}c_{\mathrm{tr}}}{\mathcal{V}_{\mathrm{tr}}c_{\mathrm{ev}}}},\qquad m^*=\frac{C-c_0}{c_{\mathrm{tr}}+c_{\mathrm{ev}}n^*}, \tag{29}$$

when both variance components and both marginal costs are positive and $C>c_0$. Substituting the budget constraint into (28) leaves the terms $\mathcal{V}_{\mathrm{tr}}c_{\mathrm{ev}}n+\mathcal{V}_{\mathrm{ev}}c_{\mathrm{tr}}/n$ as the only terms depending on n; differentiation yields (29). Zero components, integer counts and minimum sample sizes require boundary comparisons instead. Actual allocation requires estimated components and measured costs, with their uncertainty retained. This calculation supplies neither a verified CPU limit nor evidence that a particular allocation is best.

The counts may depend on earlier evidence while remaining fixed for the new experiment. Stopping the new experiment when its own accumulated cost crosses a threshold is a different sampling rule. Stochastic costs are part of the efficiency problem studied by Glynn and Whitt (1992); Appendix B gives a finite example in which an outcome-dependent cost rule biases the resulting sample mean. I retain incomplete workloads as incomplete rather than applying (28) to a selectively completed subset.

6.4 Approximation and sampling serve different purposes

A more detailed approximate value can tighten the dual expectation while making each outer path more expensive. Under a time constraint this can reduce the available sample size and increase the sampling radius in (19). Report the mean gap, its standard error, the upper limit and actual runtime separately. A finer deterministic grid is not automatically the best use of a limited evaluation budget.

For independent-method evaluations, use (22); for common-path evaluations, retain (21). Report all retained policies and repetitions. Descriptive averages or medians across related cases are not independent observations of a population effect. Neither a tighter dual construction nor a larger outer sample changes a frozen policy's expected reward. Comparing these expenditures with extra policy training is a distinct resource-allocation problem.

7 Financial validation and interpretation

A numerical stopping problem can be well defined while its market interpretation is unresolved. Empirical application requires dated contract terms, a justified ownership state, cash information available at the valuation cutoff, and funding appropriate to the payment obligations. Event and source timestamps may describe different objects. Simultaneous records can require an explicit ordering rule; absent or stale inputs must not silently become valid observations.

Joint call and put compatibility under a shared state, cash specification, funding convention and volatility is an informative diagnostic. Separate unconstrained calibrations can remove that restriction. Compatibility with bid–ask intervals is still not identification of the correct model or an executable arbitrage. A model that fails a renewed quote diagnostic after a sensitivity change can remain a conditional mathematical case, but must not be reclassified as an accepted empirical specification because its policy loss is small.

A quoted spread offers a monetary scale. It is not automatically the cost of exercise, hedging or implementation. An upper loss limit below the spread establishes only that conditional scale comparison. A limit above it leaves the inequality unestablished; it does not prove that true loss exceeds the spread.

A prospective evaluation should freeze its observations, admissibility rule, calibration and inference before outcomes can affect method choice. If no observation passes the prescribed screen, a performance estimate is unavailable. Wider post-sample thresholds can inform a future design but cannot silently replace the primary sample. Numerical replication, admissible measurement and independent economic validation are different requirements.

8 Conclusions and data availability

I distinguish policy quality from the tightness of its evaluation bound and from the financial opportunities used as the benchmark. Analytically centred values can improve the measurement of loss without changing the policy. A larger calendar changes the comparison problem, while a feasible terminal adjustment changes the rule itself. Common-upper comparisons with valid martingales identify expected policy gains; finite-family bounds and explicit timing accounting delimit what the resulting evidence can support.

I use this separation to distinguish two allocation questions. Additional computation spent on the dual construction can improve knowledge of a fixed policy's loss; additional computation spent on training can change the policy's reward. Neither expenditure can be ranked solely by the resulting dual gap because that gap combines loss and slack. Repeating fixed training procedures introduces a further distinction between variation across fitted policies and evaluation noise within a fit. The conditional variance decomposition identifies these margins without estimating their relative importance or establishing a reliable CPU forecast. An optimal allocation requires evidence about both variance and expenditure.

Predictable compensation identifies a separate source of error if the conditional mean-zero requirement is relaxed. A zero apparent gap or a common upper payoff cannot certify policy quality in its presence. I therefore retain analytical centring as a validity condition, separate from the approximation choices used to tighten the bound.

These distinctions provide a basis for a conditional computational investigation. They do not establish which specification performs best in observed markets or how the results generalise to additional state variables and uncertain funding or cash. Such conclusions require their own numerical and empirical evidence.

In this methods-only publication, I set out the model and apply the cited methods. The finite-tree example is stipulated for analytical verification. I include no raw or cleaned subscription observations, fitted market inputs, empirical summaries, research-result tables or data-derived figures. I retain the complete numerical paper separately. The LaTeX sources and build instructions for this version are available in the accompanying repository.

A Conditional drift and the interpretation of a gap

The martingale requirement is conditional. Zero average increments do not suffice. This distinction is established in the dual literature: Belomestny et al. (2009) preserve the martingale property through their construction, while Conditions A and B of Cline (2026) state conditional restrictions on projection residuals. I apply predictable compensation to express the consequence of a violation in the units of policy loss and paired policy gains. The purpose is to measure the error introduced by imperfect conditional centring. Subtracting accumulated conditional increments produces a martingale; optional sampling then isolates the part of the apparent gap that cannot be attributed to policy loss. The following argument is an elementary consequence of the cited dual framework, not a claim of a new theorem or an original proof.

A.1 A correction for an adapted penalty

Fix the trained functions and let $\mathcal{F}_j$ denote the evaluation information at exercise right j. Suppose the adapted rewards H_j and an adapted penalty A_j are integrable, $A_0 = 0$, and exercise is allowed at $1, \dots, J$. Define

$$\mu_j = \mathbb{E}[A_j - A_{j-1} \mid \mathcal{F}_{j-1}], \qquad B_j = \sum_{k=1}^{j} \mu_k, \qquad M_j = A_j - B_j, \quad B_0 = 0. \tag{30}$$

Write $V^* = \sup_\tau \mathbb{E}H_\tau$ and $V_\pi = \mathbb{E}H_{\tau_\pi}$, with the supremum taken over admissible stopping times. Then M is an integrable martingale starting at zero. For an admissible policy π, write

$$U(A) = \max_{1 \le j \le J} (H_j - A_j), \qquad G_\pi(A) = U(A) - H_{\tau_\pi} + A_{\tau_\pi}, \tag{31}$$

$$D_\pi = \max_{1 \le j \le J} B_j - B_{\tau_\pi}, \qquad \eta = \sum_{k=2}^{J} \mathbb{E}|\mu_k|. \tag{32}$$

The gap and D_π are non-negative pathwise. Nevertheless, $U(A)$ need not have an expectation above the optimal value.

Proposition 4 (Compensated loss bound)**.** *Under these assumptions,*

$$0 \le V^* - V_\pi \le \mathbb{E}[G_\pi(A) + D_\pi] \le \mathbb{E}[G_\pi(A)] + \eta. \tag{33}$$

Proof. For every admissible stopping time τ, $H_\tau \le U(A) + A_\tau$. Optional sampling applied to $M = A - B$ gives $\mathbb{E}A_\tau = \mathbb{E}B_\tau$, and likewise for τ_π. Therefore

$$\mathbb{E}H_\tau - V_\pi \le \mathbb{E}G_\pi(A) + \mathbb{E}(B_\tau - B_{\tau_\pi}) \le \mathbb{E}G_\pi(A) + \mathbb{E}D_\pi.$$

Taking the supremum over τ gives the first upper bound. The difference between any two values of B at rights $1, \dots, J$ is bounded by $\sum_{k=2}^{J} |\mu_k|$. This gives the second upper bound. Non-negativity of the loss follows from admissibility of π. □

The sum starts at two because B_1 is common to every eligible stopping right and cancels from the policy-loss comparison. If exercise at zero is included, the corresponding sum starts at one. If $\|\mu_k\|_2$ is finite, then $\mathbb{E}|\mu_k| \le \|\mu_k\|_2$ under the evaluation probability measure. Certified bounds on these norms thus suffice for this expectation bound without a uniform statewise bound. Neither finite regression residuals nor a sample estimate of a norm supplies such a certificate automatically. Division by the positive premium discount factor b converts the entire bound, including the drift allowance, into quoted premium units.

A.2 Paired policies

For two policies evaluated with the same adapted penalty, the pathwise cancellation of the upper payoff remains exact. Its expectation satisfies

$$\mathbb{E}[G_\pi(A) - G_\rho(A)] = V_\rho - V_\pi + \mathbb{E}[B_{\tau_\pi} - B_{\tau_\rho}]. \tag{34}$$

This follows by subtracting the definitions of the gaps and using $\mathbb{E}A_\tau = \mathbb{E}B_\tau$. The absolute value of the last term is at most η. Common-path evaluation therefore removes the common upper payoff but does not remove conditional-drift bias. This matters when comparing policies that stop in different states: the same residual process can be selected differently by their stopping times.

A.3 A finite analytical example

Let Z take values -1 and 1 with equal probability and become known at right one. There are two exercise rights and no additional information is needed at right two. Set $H_1 = 1$, $H_2 = 1+Z$, $A_1 = 0$ and $A_2 = Z$. These are stipulated mathematical quantities, not observations or fitted market inputs. Both unconditional increment means are zero, but $\mu_2 = Z$.

A policy π that always waits has value 1. The optimal policy stops at one when $Z = -1$ and waits otherwise, with value 3/2. Yet $U(A) = 1$ and $G_\pi(A) = 0$ in both states. Thus a non-negative, identically zero apparent gap coexists with a positive policy loss. Here $D_\pi = (-Z)^+$, so $\mathbb{E}D_\pi = 1/2$ and the first corrected bound is attained. The coarser allowance is $\eta = 1$.

The optimal policy also has a zero apparent gap in both states. Consequently the paired gap difference is zero although the true gain is 1/2. The compensator term in (34) is $-1/2$, accounting for the discrepancy. The example illustrates the conditional requirement; I make no claim of priority for this construction.

A.4 Numerical and inferential scope

For the analytically centred functions used in this paper, $\mu_j = 0$ in exact arithmetic when the stated transition and retained function are integrated exactly. The correction concerns substitutions such as approximate conditional integration or projection. It is not evidence of a defect in the retained construction, and it does not itself certify floating-point error.

If an evaluation procedure supplies an upper confidence limit for $\mathbb{E}G_\pi(A)$ and a deterministic certified bound on η, their sum has the same coverage for policy loss. If the drift allowance is estimated, its uncertainty requires separate control, for example by simultaneous limits with a declared combined error probability. An asymptotic limit for the gap remains asymptotic after the correction. Testing unconditional sample means cannot replace the conditional requirement.

The accompanying `verification/conditional_drift.py` enumerates stopping policies on finite trees using rational arithmetic. It checks the displayed example, the compensated inequality and the paired identity against an independently evaluated dynamic programme. These calculations validate the finite implementation of the argument. They neither establish theoretical novelty nor supply empirical findings.

B Training information and conditional allocation

B.1 The sampling space and the learned stopping rule

I make the probability construction explicit because a fitted stopping rule uses information acquired before its evaluation path begins. The financial specification, training procedures, their within-pair coupling and the retained approximate values are fixed throughout this construction. Let λ be the law of one complete evaluation path induced by the maintained pricing measure $\mathbb{Q}$, and let ν be the joint law of one training pair. Let the preceding record have law Π and generate $\mathcal{P}$. On standard Borel record, training and path spaces, an ideal simulation law is

$$\mathbb{P} = \Pi \otimes \bigotimes_{r\geq 1} \left(\nu \otimes \bigotimes_{i\geq 1} \lambda \right). \tag{35}$$

This construction supplies independent rows $(D_r, (X_{ri})_{i\geq 1})$ independent of the preceding record. Within a row the same fitted pair is used for every path. Conditional on $\mathcal{P}$, the finite positive counts $m = m(\mathcal{P})$ and $n = n(\mathcal{P})$ select an initial rectangle of this fresh array; they do not alter its product law. These independence statements are assumptions of the sampling model, not consequences of assigning different names to random-number streams.

Each training rule must be measurable in its input data. In a finite regression algorithm, measurable state filters, arithmetic on their declared domains, specified tie rules and finitely many safeguard branches preserve measurability. For a real design matrix A, its Moore–Penrose inverse is Borel measurable because

$$A^{+} = \lim_{\varepsilon\downarrow 0} (A^{\mathsf{T}}A + \varepsilon I)^{-1} A^{\mathsf{T}}$$

is a pointwise limit of continuous maps. A finite, explicitly specified rank or degree-selection rule therefore permits backward induction to produce jointly measurable continuation functions of training data and state. This is a requirement on the implemented rule, including its exceptional-input behaviour; it is not an assertion that an unspecified numerical routine has this property.

For one fresh path, write $\mathcal{F}_j$ for its information through exercise right j and enlarge initially to

$$\mathcal{G}_j = \mathcal{P} \vee \sigma(D) \vee \mathcal{F}_j.$$

The fitted policy is a bounded $\mathcal{G}_j$-stopping time because its parameters are initially available and its exercise predicates use no future path state. The training pair and preceding record are independent of the complete fresh path. Consequently, for each retained integrable function v_j,

$$\mathbb{E}_{\mathbb{P}}[v_j(Y_j) \mid \mathcal{G}_{j-1}] = \mathbb{E}_{\lambda}[v_j(Y_j) \mid \mathcal{F}_{j-1}]. \tag{36}$$

Thus the centred increments in (10) remain martingale increments in the information actually used by the learned rule. Integrability at the finitely many rights permits bounded-time optional sampling, conditionally on $\mathcal{P}$ and D. It gives $\mathbb{E}[M_{\tau_{\pi_a(D)}} \mid \mathcal{P}, D] = 0$ for $a = 0, 1$ and hence the expression for $\delta(D)$ below (26). I use this argument to justify the training-averaged policy comparison, applying the approximate-value construction of Haugh and Kogan (2004) to the regression policies of Longstaff and Schwartz (2001).

Equation (36) fixes the functions v_j. If the martingale functions are also trained, their inputs must be included in the initial training information and their conditional expectations must integrate the realised functions. Integrating an average function in their place need not centre the realised increment. Additional joint integrability is then required. The initial enlargement cannot simply be omitted because the parameters were fitted before evaluation.

B.2 Variance, estimation and the scope of conditioning

Under the finite-second-moment assumption for Z, paths within a row are conditionally independent given D_r with mean $\delta(D_r)$ and variance $\sigma^2_{D_r}$. Conditional total variance yields

$$\mathbb{E}[W_r \mid \mathcal{P}] = \mu, \qquad \mathrm{Var}(W_r \mid \mathcal{P}) = \mathcal{V}_{\mathrm{tr}} + \mathcal{V}_{\mathrm{ev}}/n.$$

Distinct rows are conditionally independent, which proves (28). Individual gains within a row generally are not independent after training is averaged out: two different gains have covariance $\mathcal{V}_{\mathrm{tr}}$. Treating all mn gains as independent would therefore remove the training component incorrectly.

For $m \geq 2$, let S^2_W be the usual sample variance of the m row means. Then S^2_W/m is conditionally unbiased for the variance in (28). Integrating over the preceding record gives

$$\mathrm{Var}(\widehat{\mu}) = \mathcal{V}_{\mathrm{tr}}\mathbb{E}[1/m] + \mathcal{V}_{\mathrm{ev}}\mathbb{E}[1/(mn)]. \tag{37}$$

There is no variance of a record-dependent mean because the maintained target μ is fixed. Inverse expected counts cannot replace expected inverse counts. If the preceding record changes only the coupling within each training pair while preserving both marginal fitted-policy laws, the target remains μ, but the variance components can depend on $\mathcal{P}$. Their contribution becomes $\mathbb{E}[\mathcal{V}_{\mathrm{tr}}(\mathcal{P})/m + \mathcal{V}_{\mathrm{ev}}(\mathcal{P})/(mn)]$. Selecting the financial model or marginal training procedures can additionally change the target to $\mu(\mathcal{P})$, introducing $\mathrm{Var}\{\mu(\mathcal{P})\}$ through total variance. Equation (37) assumes fixed laws and does not supply that broader uncertainty assessment.

Unbiased variance estimation is not an exact confidence statement. For each fixed positive inner count, independent rows with positive finite variance admit the usual studentised central limit approximation as their number diverges. If the inner count is chosen from a fixed finite set using only $\mathcal{P}$, the same limiting justification applies when the selected row count diverges in probability: conditional approximation errors are uniformly small across that finite set and can then be averaged over $\mathcal{P}$. This argument gives no finite-sample coverage guarantee, no assurance for an expanding count set and no licence to collapse an interval because the observed variance is zero.

B.3 Why a cost stopping rule needs a separate argument

For a finite analytical example, let independent signs ξ_i take values -1 and 1 with equal probabilities, and let the cost of observation i be $(3+\xi_i)/2$. Stop after the first completed observation for which accumulated cost reaches two. The stopping count K is one if $\xi_1 = 1$ and two otherwise. The stopped sum has expectation zero by bounded-time optional sampling, but its sample mean satisfies

$$\mathbb{E}\left[\frac{1}{K}\sum_{i=1}^{K}\xi_i\right] = \frac{1}{2}(1) + \frac{1}{4}(0) + \frac{1}{4}(-1) = \frac{1}{4}.$$

The three terms enumerate the first positive sign, a negative followed by a positive sign, and two negative signs. The example contains no observed processor timings or market data. Its purpose is to show why dividing a stopped martingale sum by its outcome-dependent count can introduce bias. I do not infer that every CPU stopping rule is biased; rather, the fixed-count proof supplies no justification for such a rule. The example and the preceding derivations are applications of conditional expectation and optional sampling, without a claim of theoretical priority.

The accompanying `verification/allocation_probability.py` uses exact rational arithmetic on stipulated finite sampling laws to check the conditional variance identities and this stopping example. These finite calculations check an implementation of the argument; they do not verify the full measurable-space construction, certify a random-number generator or provide empirical evidence.

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